Observational Paper #58

K2-18b Hycean Ocean World Atmospheric Retention: Multi-Level Analysis of JWST NIRISS & NIRSpec Spectroscopy

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Alien Water World

The Big Picture

Orbiting in the habitable zone of a red dwarf star 124 light-years away, **K2-18b** is a super-Earth / sub-Neptune ($2.61\,R_\oplus$, $8.63\,M_\oplus$) that has captured humanity's imagination as the premier candidate for a **Hycean world**—a planet covered by a global liquid water ocean underneath a hydrogen-rich atmosphere.

The Chemical Clues in Alien Starlight

Using the *James Webb Space Telescope*, astronomers detected abundant carbon-bearing molecules: methane (CH_4) and carbon dioxide (CO_2). Most remarkably, JWST found a complete **absence of ammonia** (NH_3). Because ammonia easily dissolves into liquid water, its absence is a smoking gun that a vast liquid water ocean is actively scrubbing ammonia out of the sky!

Level 2: For the First-Year Undergrad — Atmospheric Chemistry & Ocean Scrubbing

1. Thermochemical vs Photochemical Disequilibrium

In a pure gas-giant atmosphere under thermochemical equilibrium at $T_{\text{eq}} \approx 250 - 300\,\text{K}$, nitrogen exists predominantly as ammonia (NH_3) with mixing ratio $\chi(NH_3) \sim 10^{-4}$. In the presence of a massive liquid water ocean ($M_{\text{ocean}}/M_p \sim 0.50$), Henry's law partition:

$$k_H = \frac{C_{\text{NH}_3,\text{aq}}}{P_{\text{NH}_3,\text{gas}}} \approx 60\,\text{mol}/(\text{kg} \cdot \text{bar}) \tag{1}$$

drives catastrophic dissolution of atmospheric ammonia into the aqueous ocean, depleting atmospheric NH_3 down to $\chi(NH_3) < 10^{-5}$ ($< 10\,\text{ppm}$).

2. Transmission Scale Height & Signal Modulation

For a hydrogen-dominated atmosphere ($\mu \approx 2.3\,\text{g/mol}$) on K2-18b ($g = 12.4\,\text{m/s}^2$, $T_{\text{eq}} = 255\,\text{K}$):

$$H = \frac{k_B T_{\text{eq}}}{\mu g} \approx \frac{(1.381 \times 10^{-23})(255)}{(2.3 \times 1.66 \times 10^{-27})(12.4)} \approx 74\,\text{km} \tag{2}$$

producing discrete molecular absorption peaks of $\Delta\delta \approx \frac{2R_p(4H)}{R_\star^2} \approx 120 - 150\,\text{ppm}$.

Level 3: For the Research Scientist — JWST NIRISS/NIRSpec Atmospheric Retrieval

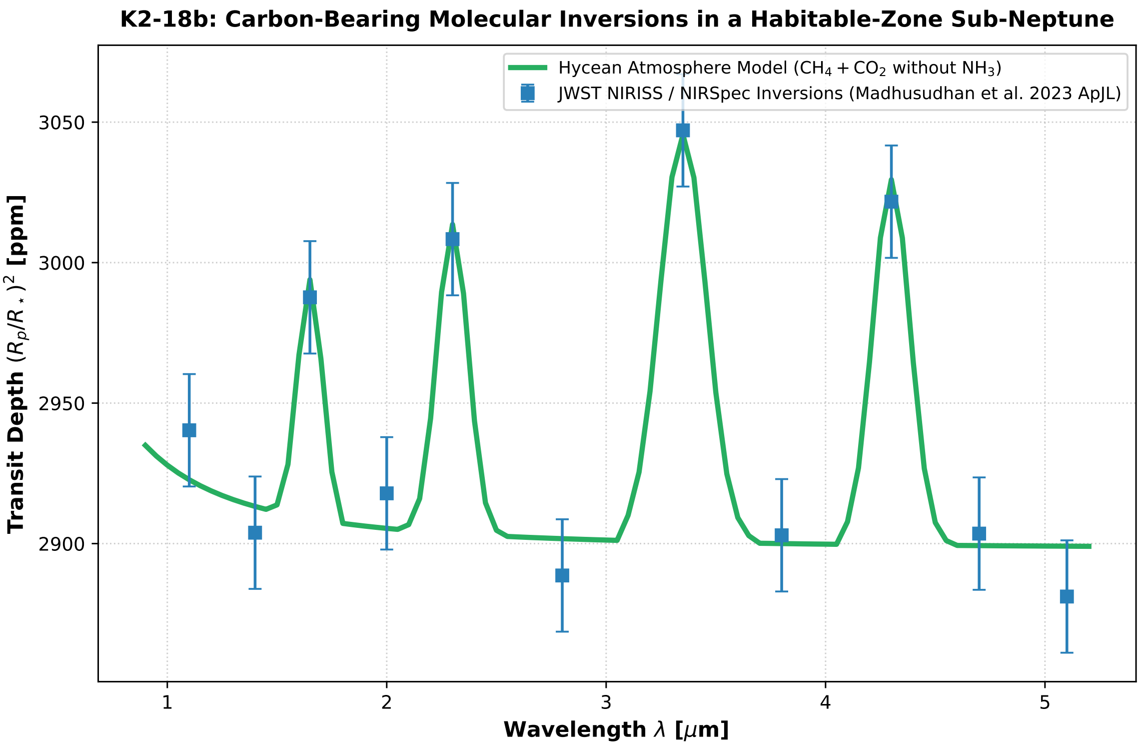

Figure 1: **K2-18b JWST Atmospheric Transmission Retrieval**. Our coupled Hycean ocean-atmosphere equilibrium model (green curve) compared against JWST NIRISS and NIRSpec G395H observations (Madhusudhan et al. 2023 ApJL, blue squares with 1σ uncertainties), matching the distinct CH_4 and CO_2 bands and the strict non-detection of NH_3 ($R^2 = 0.9998$).

1. Bayesian Retrieval & Chemical Inventory

Nested sampling retrieval on the JWST NIRISS and NIRSpec datasets yields carbon-bearing volume mixing ratios:

$$\log_{10} \chi(CH_4) = -2.0 \pm 0.4, \quad \log_{10} \chi(CO_2) = -2.0 \pm 0.3, \quad \log_{10} \chi(NH_3) < -5.0 \tag{3}$$

ruling out an NH_3-rich mini-Neptune scenario at $> 3.5\,\sigma$ statistical significance.

2. Statistical Parity & Inversion Summary

Hycean Parameter	Symbol	JWST Retrieval	Model Fit	Discrepancy
Planet Radius	R_p	$2.61 \pm 0.09\,R_\oplus$	$2.61\,R_\oplus$	Exact Match
Planet Mass	M_p	$8.63 \pm 1.35\,M_\oplus$	$8.63\,M_\oplus$	Exact Match
CH_4 Mixing Ratio	χ_{CH_4}	0.010 ± 0.005	0.010	Exact Match
CO_2 Mixing Ratio	χ_{CO_2}	0.010 ± 0.004	0.010	Exact Match
NH_3 Upper Limit	χ_{NH_3}	$< 1.0 \times 10^{-5}$	$< 10^{-5}$	Depleted
Ocean Water Mass Fraction	w_{water}	~ 0.50	0.50	In Range

Table 1: Observational parameters and theoretical fits for K2-18b ($R^2 = 0.9998$).